

Multiple Reentry Vehicle Terminal Precision, Hypersonic Ablation Dynamics, and Guidance Synthesis

Advanced Ballistic Dynamics & Exo-/Endo-Atmospheric Seminar

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Problem Statement

Consider an exo-atmospheric separation of a multiple independently targetable reentry vehicle (MIRV) payload at high altitude ($z \geq 100$ km). Following post-boost vehicle detachment, an individual maneuvering reentry vehicle (MaRV) with time-varying mass $m(t)$ enters the terminal dense atmospheric region ($z \leq 30$ km) targeting a precise high-value ground facility.

The atmosphere exhibits non-isothermal exponential stratification:

$$\rho(z) = \rho_0 \exp\left(-\frac{z}{H}\right), \quad T(z) = T_0 - \lambda z \quad (1)$$

where ρ_0 is sea-level air density, H is atmospheric scale height, T_0 is sea-level temperature, and λ is lapse rate[cite: 1]. The warhead employs an integrated closed-loop guidance and control strategy to suppress wind perturbations, offset thermal-ablation dynamic imbalances, and ensure zero circular error probable (CEP) at ground impact.

Part A: MIRV Terminal Kinematics, Lie Groups, and Frame Rotations

Question 1: Quaternion Kinematics & Attitude Invariance under Terminal Entry

Let $q = [q_0 \ q_1 \ q_2 \ q_3]^T \in \mathbb{S}^3$ represent the orientation quaternion mapping vectors from the local earth-centered inertial frame I to the warhead body frame B [cite: 1]. The vehicle experiences rapid angular body rates $\omega_b = [\omega_x \ \omega_y \ \omega_z]^T$ during atmospheric entry trim setup[cite: 1].

- (a) Derive the complete differential kinematic relation $\dot{q} = \frac{1}{2}\Omega(\omega_b)q$, explicitly constructing the 4×4 skew-symmetric angular rate matrix $\Omega(\omega_b)$ [cite: 1].
- (b) Prove rigorously that the unit norm constraint $\|q(t)\|_2^2 = 1$ is preserved for all terminal time $t \geq 0$ under non-zero body rotation rates[cite: 1].

Question 2: Target Line-of-Sight Kinematics for Terminal Precision

Let $r(t) = r_{\text{target}}(t) - r_{\text{warhead}}(t) \in \mathbb{R}^3$ be the instantaneous relative range vector to the ground target, $R(t) = \|r(t)\|_2$ be the range, and $u(t) = \frac{r(t)}{R(t)} \in \mathbb{S}^2$ be the Line-of-Sight (LOS) unit vector[cite: 1].

- (a) Prove that the target LOS rotation rate vector Ω_{LOS} satisfies $\Omega_{\text{LOS}} = u \times \dot{u} = \frac{r \times v_{\text{rel}}}{R^2}$, where $v_{\text{rel}} = \dot{r}$ [cite: 1].
- (b) Derive the exact vector expression for the second time derivative $\ddot{u}$ as a function of range R , range rate $\dot{R}$, Ω_{LOS} , and relative acceleration vector $a_{\text{rel}} = \ddot{r}$ [cite: 1].

Part B: Hypersonic Aerodynamics, Thermal Ablation, & Path Energy

Question 3: Aerodynamic Frame Transformations & Angle-of-Attack Sensitivity

The hypersonic entry forces in the wind frame W are governed by[cite: 1]:

$$F_w = \begin{bmatrix} -D \\ Y \\ -L \end{bmatrix} = \frac{1}{2} \rho(z) \|v_v\|_2^2 S_{\text{ref}} \begin{bmatrix} -(C_{D0} + K\alpha^2 + K_\beta\beta^2) \\ C_{Y\beta}\beta \\ -(C_{L0} + C_{L\alpha}\alpha) \end{bmatrix} \quad (2)$$

where $\alpha = \arctan\left(\frac{v_{bz}}{v_{bx}}\right)$ and $\beta = \arcsin\left(\frac{v_{by}}{\|v_v\|}\right)$ [cite: 1].

- (a) Evaluate the local linear Jacobian matrix $J_{\alpha,\beta} = \frac{\partial(\alpha,\beta)}{\partial v_b} \in \mathbb{R}^{2 \times 3}$ mapping body velocity variations $v_b = [v_{bx} \ v_{by} \ v_{bz}]^T$ to aerodynamic orientation angles (α, β) at symmetric trim flight $(\alpha_0, \beta_0 = 0)$ [cite: 1].

Question 4: Thermal Energy Absorption during Terminal Atmospheric Penetration

Convective nose-cone heating on the carbon-carbon ablation nose cap of radius R_n is modeled via the Fay-Riddell equation $q_s(t) = C_H \sqrt{\frac{\rho(z(t))}{R_n}} \|v_v(t)\|_2^3$ [cite: 1].

- (a) Assuming a steep terminal entry trajectory where $z(t) = z_0 - (v_0 \sin \gamma_0)t$ and hypersonic velocity decays due to atmospheric drag as $\|v_v(t)\|_2 = v_0 e^{-kt}$ [cite: 1], evaluate the total accumulated thermal energy per unit surface area $Q_{\text{total}} = \int_0^{T_f} q_s(t) dt$ in closed integrated form[cite: 1].

Part C: Terminal Guidance Linear State-Space Stability & Pole Assignment

Question 5: Coupled Pitch-Yaw Reentry Dynamics & Closed-Loop Control

The perturbed dynamic linearized state equations of the reentry vehicle during terminal descent are represented as $\dot{x}(t) = Ax(t) + Bu_c(t)$ with $x = [\Delta\alpha \quad \Delta q \quad \Delta\beta \quad \Delta r]^T \in \mathbb{R}^4$ [cite: 1]:

$$A = \begin{bmatrix} -\frac{Z_\alpha}{V_0} & 1 & 0 & 0 \\ M_\alpha & M_q & 0 & 0 \\ 0 & 0 & \frac{Y_\beta}{V_0} & -1 \\ 0 & 0 & N_\beta & N_r \end{bmatrix}, \quad B = \begin{bmatrix} -\frac{Z_\delta}{V_0} & 0 \\ M_\delta & 0 \\ 0 & \frac{Y_\delta}{V_0} \\ 0 & N_\delta \end{bmatrix} \quad (3)$$

- (a) Compute the full 4th-order system characteristic polynomial $P(\lambda) = \det(\lambda I - A) = 0$ using block matrix determinant expansion[cite: 1].
- (b) Apply the Routh-Hurwitz criterion to the longitudinal pitch block A_{11} to determine necessary and sufficient conditions on aerodynamic derivatives $(Z_\alpha, M_\alpha, M_q)$ for intrinsic dynamic stability ($\text{Re}(\lambda_i) < 0$)[cite: 1].
- (c) Construct the full state feedback pole-placement algebra for control matrix $K \in \mathbb{R}^{2 \times 4}$ under the control law $u_c = -Kx$ to place closed-loop poles at prescribed stable roots $\sigma(A - BK) = \{\mu_1, \mu_2, \mu_3, \mu_4\} \subset \mathbb{C}^-$ [cite: 1].

Part D: Strategic Mindmap & Mental Framework

Meta-Strategy: Operational Deconstruction of Terminal MIRV Precision

When solving terminal precision problems for hypersonic reentry warheads, break the physical system down into three distinct operational domains[cite: 1]:

1. **Kinematic Subsystem (Part A):** Map continuous 3D rotations from inertial space into body axes without numerical gimbal lock, and formulate the vector geometry connecting the warhead to the ground target[cite: 1].
2. **Aerothermodynamic Subsystem (Part B):** Quantify angle-of-attack sensitivities during atmospheric entry and evaluate total thermal exposure along the diving trajectory[cite: 1].
3. **Control Systems Subsystem (Part C):** Decouple state variables into planar dynamics, analyze unforced airframe stability, and design closed-loop state feedback to eliminate target tracking error[cite: 1].

1. Mindmap for Part A: MIRV Terminal Kinematics

Problem 1: Quaternion Rotations & Invariance[cite: 1]

- **Physical Context:** Tracking high-speed rotational corrections during atmospheric entry[cite: 1].
- **Core Technique:** Matrix conversion of quaternion products $\dot{q} = \frac{1}{2}q \otimes \bar{\omega}_b$ into $\dot{q} = \frac{1}{2}\Omega(\omega_b)q$ [cite: 1].
- **Mathematical Strategy:** Prove $\frac{d}{dt}(q^T q) = 0$ by leveraging the anti-symmetry property $x^T M x = 0$ for skew-symmetric matrices $\Omega^T = -\Omega$ [cite: 1].

Problem 2: Target Line-of-Sight Geometry[cite: 1]

- **Physical Context:** Terminal optical/radar target tracking from a hypersonic warhead[cite: 1].
- **Core Technique:** Vector differentiation of relative position $r = Ru$ [cite: 1].
- **Mathematical Strategy:** Cross u into relative velocity v_{rel} to isolate $\Omega_{\text{LOS}} = u \times \dot{u}$, then apply vector product/quotient differentiation rules to extract $\ddot{u}$ [cite: 1].

2. Mindmap for Part B: Aerothermodynamics & Energy Integrals

Problem 3: Velocity-to-Angle Jacobian Sensitivity[cite: 1]

- **Physical Context:** Sensitivity of aerodynamic trim angles (α, β) to body-axis wind gusts[cite: 1].
- **Core Technique:** Partial derivative evaluation of arctan and arcsin vector fields[cite: 1].
- **Mathematical Strategy:** Differentiate component-wise with respect to (v_{bx}, v_{by}, v_{bz}) and evaluate at symmetric equilibrium condition $v_{by} = 0$ [cite: 1].

Problem 4: Terminal Aerothermal Heat Flux Accumulation[cite: 1]

- **Physical Context:** Heat shield degradation and ablation structural load calculation[cite: 1].
- **Core Technique:** Parameterized definite path integration of coupled exponential fields[cite: 1].
- **Mathematical Strategy:** Substitute trajectory time functions $z(t)$ and $v(t)$ into $q_s(t)$, group exponential powers into a unified exponent $\exp(At)$, and integrate over descent time $[0, T_f]$ [cite: 1].

3. Mindmap for Part C: Closed-Loop Guidance Synthesis

Problem 5: Subsystem Decoupling, Stability, & Pole Assignment[cite: 1]

- **Physical Context:** Autopilot design for high-precision terminal divert maneuvers[cite: 1].
- **Core Technique:** Block diagonal matrix expansion, Routh-Hurwitz stability criterion, and matrix pole placement[cite: 1].
- **Mathematical Strategy:** Factor $\det(\lambda I - A)$ into longitudinal and lateral quadratic factors[cite: 1]. Apply Routh conditions $a_1 > 0, a_0 > 0$ to express stability boundaries in terms of aerodynamic derivatives[cite: 1]. Compute closed-loop matrix $(A - BK)$ and solve 2×2 algebraic equations for gain matrix K [cite: 1].

Part E: Complete Step-by-Step Mathematical Solutions

Problem 1: Quaternion Kinematics & Norm Invariance Proof

(a) Derivation of $\dot{q} = \frac{1}{2}\Omega(\omega_b)q$

The orientation quaternion is defined as[cite: 1]:

$$q = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \quad (4)$$

Under angular velocity $\omega_b = [\omega_x, \omega_y, \omega_z]^T$, the dynamic rate of change is[cite: 1]:

$$\dot{q} = \frac{1}{2}q \otimes \bar{\omega}_b \quad (5)$$

where $\bar{\omega}_b = [0, \omega_x, \omega_y, \omega_z]^T$ [cite: 1]. Applying the quaternion multiplication rule $p \otimes s = \begin{bmatrix} p_0 s_0 - p_v \cdot s_v \\ p_0 s_v + s_0 p_v + p_v \times s_v \end{bmatrix}$ [cite: 1]:

1. Scalar Part:

$$\dot{q}_0 = \frac{1}{2}(0 - (q_1\omega_x + q_2\omega_y + q_3\omega_z)) = \frac{1}{2}(-\omega_x q_1 - \omega_y q_2 - \omega_z q_3) \quad (6)$$

2. Vector Part:

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} \left(q_0 \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} + \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \times \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \right) \quad (7)$$

Evaluating the vector cross product explicitly[cite: 1]:

$$q_v \times \omega_b = \begin{bmatrix} q_2\omega_z - q_3\omega_y \\ q_3\omega_x - q_1\omega_z \\ q_1\omega_y - q_2\omega_x \end{bmatrix} \quad (8)$$

Expanding into scalar differential components[cite: 1]:

$$\dot{q}_0 = \frac{1}{2}(0 \cdot q_0 - \omega_x q_1 - \omega_y q_2 - \omega_z q_3) \quad (9)$$

$$\dot{q}_1 = \frac{1}{2}(\omega_x q_0 + 0 \cdot q_1 + \omega_z q_2 - \omega_y q_3) \quad (10)$$

$$\dot{q}_2 = \frac{1}{2}(\omega_y q_0 - \omega_z q_1 + 0 \cdot q_2 + \omega_x q_3) \quad (11)$$

$$\dot{q}_3 = \frac{1}{2}(\omega_z q_0 + \omega_y q_1 - \omega_x q_2 + 0 \cdot q_3) \quad (12)$$

Assembling into vector-matrix product form $\dot{q} = \frac{1}{2}\Omega(\omega_b)q$ yields[cite: 1]:

$$\Omega(\omega_b) = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix} \quad (13)$$

(b) Proof of Norm Invariance $\|q(t)\|_2^2 = 1$

Evaluate the time derivative of the squared norm[cite: 1]:

$$\frac{d}{dt}\|q(t)\|_2^2 = \frac{d}{dt}(q^T q) = \dot{q}^T q + q^T \dot{q} \quad (14)$$

Substitute $\dot{q} = \frac{1}{2}\Omega q$ and $\dot{q}^T = \frac{1}{2}q^T\Omega^T$ [cite: 1]:

$$\frac{d}{dt}(q^T q) = \frac{1}{2}q^T\Omega^T q + \frac{1}{2}q^T\Omega q = \frac{1}{2}q^T(\Omega^T + \Omega)q \quad (15)$$

Checking matrix skew-symmetry ($\Omega^T = -\Omega$)[cite: 1]:

$$\Omega^T = \begin{bmatrix} 0 & \omega_x & \omega_y & \omega_z \\ -\omega_x & 0 & -\omega_z & \omega_y \\ -\omega_y & \omega_z & 0 & -\omega_x \\ -\omega_z & -\omega_y & \omega_x & 0 \end{bmatrix} = -\Omega \implies \Omega^T + \Omega = 0_{4 \times 4} \quad (16)$$

Thus[cite: 1]:

$$\frac{d}{dt}\|q(t)\|_2^2 = \frac{1}{2}q^T(0_{4 \times 4})q = 0 \quad (17)$$

Integrating from $t = 0$ with unit initial conditions $\|q(0)\|_2 = 1$ confirms $\|q(t)\|_2^2 = 1$ is invariant for all time $t \geq 0$ [cite: 1].

Problem 2: Target Line-of-Sight Kinematics Vector Derivations

(a) Proof of $\Omega_{\text{LOS}} = u \times \dot{u} = \frac{r \times v_{\text{rel}}}{R^2}$

From relative geometry $r = Ru$, differentiate with respect to time[cite: 1]:

$$v_{\text{rel}} = \dot{r} = \dot{R}u + R\dot{u} \quad (18)$$

Take the cross product from the left with u [cite: 1]:

$$u \times v_{\text{rel}} = u \times (\dot{R}u + R\dot{u}) = \dot{R}(u \times u) + R(u \times \dot{u}) \quad (19)$$

Since $u \times u = 0$ [cite: 1]:

$$u \times v_{\text{rel}} = R(u \times \dot{u}) \implies u \times \dot{u} = \frac{u \times v_{\text{rel}}}{R} \quad (20)$$

Substitute $u = \frac{r}{R}$ [cite: 1]:

$$u \times \dot{u} = \frac{\left(\frac{r}{R}\right) \times v_{\text{rel}}}{R} = \frac{r \times v_{\text{rel}}}{R^2} \quad (21)$$

Since $\Omega_{\text{LOS}} = u \times \dot{u}$, this completes the proof[cite: 1]:

$$\Omega_{\text{LOS}} = \frac{r \times v_{\text{rel}}}{R^2} \quad (22)$$

(b) Derivation of Second Derivative $\ddot{u}$

Isolate $\dot{u}$ from relative velocity $v_{\text{rel}} = \dot{R}u + R\dot{u}$ [cite: 1]:

$$\dot{u} = \frac{v_{\text{rel}} - \dot{R}u}{R} \quad (23)$$

Differentiate using the quotient rule[cite: 1]:

$$\ddot{u} = \frac{R \frac{d}{dt}(v_{\text{rel}} - \dot{R}u) - (v_{\text{rel}} - \dot{R}u)\dot{R}}{R^2} \quad (24)$$

Compute $\frac{d}{dt}(v_{\text{rel}} - \dot{R}u) = a_{\text{rel}} - \ddot{R}u - \dot{R}\dot{u}$ [cite: 1]:

$$\ddot{u} = \frac{R(a_{\text{rel}} - \ddot{R}u - \dot{R}\dot{u}) - \dot{R}(\dot{R}u + R\dot{u}) + \dot{R}^2 u}{R^2} \quad (25)$$

$$= \frac{Ra_{\text{rel}} - R\ddot{R}u - 2R\dot{R}\dot{u} - \dot{R}^2 u + \dot{R}^2 u}{R^2} \quad (26)$$

$$= \frac{a_{\text{rel}} - \ddot{R}u - 2\dot{R}\dot{u}}{R} \quad (27)$$

Using vector identity $\dot{u} = \Omega_{\text{LOS}} \times u$, substitute into the acceleration expression[cite: 1]:

$$\ddot{u} = \frac{a_{\text{rel}} - \ddot{R}u - 2\dot{R}(\Omega_{\text{LOS}} \times u)}{R} \quad (28)$$

Problem 3: Aerodynamic Frame Transformation & Jacobian Evaluation

The aerodynamic incident angles are[cite: 1]:

$$\alpha(v_b) = \arctan\left(\frac{v_{bz}}{v_{bx}}\right), \quad \beta(v_b) = \arcsin\left(\frac{v_{by}}{V}\right) \quad (29)$$

where $V = \sqrt{v_{bx}^2 + v_{by}^2 + v_{bz}^2}$ [cite: 1].

1. Partial Derivatives of Angle of Attack α

$$\frac{\partial \alpha}{\partial v_{bx}} = \frac{1}{1 + \left(\frac{v_{bz}}{v_{bx}}\right)^2} \left(-\frac{v_{bz}}{v_{bx}^2}\right) = -\frac{v_{bz}}{v_{bx}^2 + v_{bz}^2} \quad (30)$$

$$\frac{\partial \alpha}{\partial v_{by}} = 0 \quad (31)$$

$$\frac{\partial \alpha}{\partial v_{bz}} = \frac{1}{1 + \left(\frac{v_{bz}}{v_{bx}}\right)^2} \left(\frac{1}{v_{bx}}\right) = \frac{v_{bx}}{v_{bx}^2 + v_{bz}^2} \quad (32)$$

At symmetric equilibrium flight where $v_{by} = 0$, $v_{bx} = V_0 \cos \alpha_0$ and $v_{bz} = V_0 \sin \alpha_0$ [cite: 1]:

$$\frac{\partial \alpha}{\partial v_{bx}} = -\frac{\sin \alpha_0}{V_0}, \quad \frac{\partial \alpha}{\partial v_{by}} = 0, \quad \frac{\partial \alpha}{\partial v_{bz}} = \frac{\cos \alpha_0}{V_0} \quad (33)$$

2. Partial Derivatives of Sideslip Angle β

Applying $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$ [cite: 1]:

$$\frac{\partial \beta}{\partial v_i} = \frac{1}{\sqrt{1 - \left(\frac{v_{by}}{V}\right)^2}} \frac{\partial}{\partial v_i} \left(\frac{v_{by}}{V}\right) \quad (34)$$

At trim flight ($\beta_0 = 0 \implies v_{by} = 0$), the pre-factor reduces to 1[cite: 1]. Differentiating $f = \frac{v_{by}}{V}$ [cite: 1]:

$$\left. \frac{\partial \beta}{\partial v_{bx}} \right|_{\text{trim}} = 0 \quad (35)$$

$$\left. \frac{\partial \beta}{\partial v_{by}} \right|_{\text{trim}} = \left. \frac{V(1) - v_{by} \frac{v_{by}}{V}}{V^2} \right|_{v_{by}=0} = \frac{V_0}{V_0^2} = \frac{1}{V_0} \quad (36)$$

$$\left. \frac{\partial \beta}{\partial v_{bz}} \right|_{\text{trim}} = 0 \quad (37)$$

3. Assembled Linearized Jacobian Matrix

Combining partial derivative rows into matrix form[cite: 1]:

$$J_{\alpha,\beta}|_{\text{trim}} = \begin{bmatrix} -\frac{\sin \alpha_0}{V_0} & 0 & \frac{\cos \alpha_0}{V_0} \\ 0 & \frac{1}{V_0} & 0 \end{bmatrix} \quad (38)$$

Problem 4: Closed-Form Aerothermal Integral Computation

The stagnation point convective heat flux formula is[cite: 1]:

$$q_s(t) = C_H \sqrt{\frac{\rho(z(t))}{R_n}} \|v_v(t)\|_2^3 \quad (39)$$

Given altitude trajectory $z(t) = z_0 - (v_0 \sin \gamma_0)t$, atmospheric density decay $\rho(z) = \rho_0 e^{-z/H}$, and drag velocity profile $\|v_v(t)\|_2 = v_0 e^{-kt}$ [cite: 1]:

1. Substitution of State Functions into Heat Flux Integrand

Evaluate air density square root along path[cite: 1]:

$$\sqrt{\rho(z(t))} = \sqrt{\rho_0 \exp\left(-\frac{z_0 - (v_0 \sin \gamma_0)t}{H}\right)} = \sqrt{\rho_0} \exp\left(-\frac{z_0}{2H}\right) \exp\left(\frac{v_0 \sin \gamma_0}{2H}t\right) \quad (40)$$

Evaluate cubed velocity profile[cite: 1]:

$$\|v_v(t)\|_2^3 = (v_0 e^{-kt})^3 = v_0^3 e^{-3kt} \quad (41)$$

Combining into $q_s(t)$ [cite: 1]:

$$q_s(t) = C_H v_0^3 \sqrt{\frac{\rho_0}{R_n}} \exp\left(-\frac{z_0}{2H}\right) \exp\left[\left(\frac{v_0 \sin \gamma_0}{2H} - 3k\right)t\right] \quad (42)$$

2. Definite Path Integration

Define constants $K_{\text{const}} = C_H v_0^3 \sqrt{\frac{\rho_0}{R_n}} e^{-\frac{z_0}{2H}}$ and unified power constant $A = \frac{v_0 \sin \gamma_0}{2H} - 3k$ [cite: 1]. Total thermal energy per unit area is[cite: 1]:

$$Q_{\text{total}} = \int_0^{T_f} q_s(t) dt = K_{\text{const}} \int_0^{T_f} e^{At} dt = \frac{K_{\text{const}}}{A} (e^{AT_f} - 1) \quad (43)$$

Substituting full physical parameters yields the exact closed form[cite: 1]:

$$Q_{\text{total}} = \frac{C_H v_0^3 \sqrt{\frac{\rho_0}{R_n}} \exp\left(-\frac{z_0}{2H}\right)}{\frac{v_0 \sin \gamma_0}{2H} - 3k} \left[\exp\left(\left[\frac{v_0 \sin \gamma_0}{2H} - 3k\right] T_f\right) - 1 \right] \quad (44)$$

Problem 5: Linear System Stability & Control Synthesis

(a) System Characteristic Polynomial $P(\lambda)$

The 4×4 system matrix A is block diagonal[cite: 1]:

$$A = \begin{bmatrix} A_{11} & 0_{2 \times 2} \\ 0_{2 \times 2} & A_{22} \end{bmatrix} = \begin{bmatrix} -\frac{Z_\alpha}{V_0} & 1 & 0 & 0 \\ M_\alpha & M_q & 0 & 0 \\ 0 & 0 & \frac{Y_\beta}{V_0} & -1 \\ 0 & 0 & N_\beta & N_r \end{bmatrix} \quad (45)$$

Applying the block determinant property $\det(A) = \det(A_{11}) \det(A_{22})$ [cite: 1]:

$$P(\lambda) = \det(\lambda I_4 - A) = \det(\lambda I_2 - A_{11}) \cdot \det(\lambda I_2 - A_{22}) \quad (46)$$

1. Pitch Subsystem Determinant:

$$\det(\lambda I_2 - A_{11}) = \det \begin{bmatrix} \lambda + \frac{Z_\alpha}{V_0} & -1 \\ -M_\alpha & \lambda - M_q \end{bmatrix} = \lambda^2 + \left(\frac{Z_\alpha}{V_0} - M_q\right) \lambda + \left(-\frac{Z_\alpha M_q}{V_0} - M_\alpha\right) \quad (47)$$

2. Yaw Subsystem Determinant:

$$\det(\lambda I_2 - A_{22}) = \det \begin{bmatrix} \lambda - \frac{Y_\beta}{V_0} & 1 \\ -N_\beta & \lambda - N_r \end{bmatrix} = \lambda^2 - \left(\frac{Y_\beta}{V_0} + N_r\right) \lambda + \left(\frac{Y_\beta N_r}{V_0} + N_\beta\right) \quad (48)$$

Multiplying both sub-polynomials gives $P(\lambda) = 0$ [cite: 1]:

$$P(\lambda) = \left[\lambda^2 + \left(\frac{Z_\alpha}{V_0} - M_q\right) \lambda - \left(\frac{Z_\alpha M_q}{V_0} + M_\alpha\right) \right] \cdot \left[\lambda^2 - \left(\frac{Y_\beta}{V_0} + N_r\right) \lambda + \left(\frac{Y_\beta N_r}{V_0} + N_\beta\right) \right] \quad (49)$$

(b) Routh-Hurwitz Stability Boundaries for Pitch Subsystem

For quadratic polynomial $P_{\text{pitch}}(\lambda) = \lambda^2 + a_1\lambda + a_0 = 0$, asymptotic stability ($\text{Re}(\lambda_i) < 0$) requires $a_1 > 0$ and $a_0 > 0$ [cite: 1]:

$$a_1 = \frac{Z_\alpha}{V_0} - M_q > 0 \implies M_q < \frac{Z_\alpha}{V_0} \quad (50)$$

$$a_0 = -\frac{Z_\alpha M_q}{V_0} - M_\alpha > 0 \implies M_\alpha < -\frac{Z_\alpha M_q}{V_0} \quad (51)$$

Because lift curve slope $Z_\alpha > 0$ and damping derivative $M_q < 0$, the stability bounds require restoring pitching moment derivative $M_\alpha < 0$ [cite: 1].

(c) Full State Feedback Pole-Placement Control Matrix Synthesis

Let control input be $u_c = -Kx$ with $K = \begin{bmatrix} k_{11} & k_{12} & 0 & 0 \\ 0 & 0 & k_{23} & k_{24} \end{bmatrix}$ [cite: 1]. Compute closed-loop matrix product BK [cite: 1]:

$$BK = \begin{bmatrix} -\frac{Z_\delta k_{11}}{V_0} & -\frac{Z_\delta k_{12}}{V_0} & 0 & 0 \\ M_\delta k_{11} & M_\delta k_{12} & 0 & 0 \\ 0 & 0 & \frac{Y_\delta k_{23}}{V_0} & \frac{Y_\delta k_{24}}{V_0} \\ 0 & 0 & N_\delta k_{23} & N_\delta k_{24} \end{bmatrix} \quad (52)$$

The closed-loop longitudinal matrix $A_{\text{cl},11} = A_{11} - (BK)_{11}$ is[cite: 1]:

$$A_{\text{cl},11} = \begin{bmatrix} -\frac{Z_\alpha - Z_\delta k_{11}}{V_0} & 1 + \frac{Z_\delta k_{12}}{V_0} \\ M_\alpha - M_\delta k_{11} & M_q - M_\delta k_{12} \end{bmatrix} \quad (53)$$

The characteristic polynomial of $A_{\text{cl},11}$ is $\lambda^2 + \bar{a}_1\lambda + \bar{a}_0$, where[cite: 1]:

$$\bar{a}_1(k_{11}, k_{12}) = \frac{Z_\alpha - Z_\delta k_{11}}{V_0} - (M_q - M_\delta k_{12}) \quad (54)$$

$$\bar{a}_0(k_{11}, k_{12}) = \left(\frac{Z_\alpha - Z_\delta k_{11}}{V_0} \right) (M_q - M_\delta k_{12}) - \left(1 + \frac{Z_\delta k_{12}}{V_0} \right) (M_\alpha - M_\delta k_{11}) \quad (55)$$

Equating coefficients to the target polynomial $P_{\text{target}}(\lambda) = (\lambda - \mu_1)(\lambda - \mu_2) = \lambda^2 - (\mu_1 + \mu_2)\lambda + \mu_1\mu_2$ yields a solvable 2×2 algebraic linear system for gains k_{11} and k_{12} [cite: 1]:

$$\bar{a}_1(k_{11}, k_{12}) = -(\mu_1 + \mu_2) \quad (56)$$

$$\bar{a}_0(k_{11}, k_{12}) = \mu_1\mu_2 \quad (57)$$

Applying the equivalent matching procedure to $A_{\text{cl},22}$ with target poles (μ_3, μ_4) uniquely specifies lateral gains k_{23} and k_{24} [cite: 1].